

Healthcare Dataset Analysis

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Calculating Basic Statistics for Numerical Variables

Introduction

This analysis examines a mock healthcare dataset containing patient information from outpatient clinic visits. The dataset includes 5 patients with various demographic and health metrics. The primary objective is to calculate and explain the basic statistics for two key numerical variables: Age and Heart Rate. Statistical measures such as the mean (average) are fundamental tools in healthcare analytics for understanding patient populations and identifying trends (Dawson & Trapp, 2004).

Dataset Overview

The dataset contains the following five patient records with 15 variables including demographics, visit information, and clinical measurements:

Patient ID	Age	Gender	Visit Type	Heart Rate (bpm)	BMI
1001	45	Female	Regular Checkup	72	24.5
1002	62	Male	Follow-up	78	29.2
1003	28	Female	Urgent Care	88	22.1
1004	55	Male	Regular Checkup	76	31.5

1005	39	Female	Follow-up	70	26.8
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Methodology

The arithmetic mean (average) is calculated using the standard formula for descriptive statistics (Moore, Notz, & Fligner, 2013):

$$\text{Mean } (\bar{x}) = \Sigma x / n$$

Where Σx represents the sum of all values and n represents the number of observations. This measure of central tendency provides a single value that represents the typical value in the dataset (Pagano & Gauvreau, 2000).

Calculation 1: Average Age

Step 1: Identify all age values from the dataset

From the five patient records, I extract the age values:

Patient 1001: 45 years Patient 1002: 62 years Patient 1003: 28 years Patient 1004: 55 years Patient 1005: 39 years

Step 2: Add all age values together

Sum of ages (Σx) = 45 + 62 + 28 + 55 + 39

= 107 + 28 + 55 + 39 (adding first two numbers) = 135 + 55 + 39 (adding next number) = 190 + 39 (adding next number) = 229 years (final sum)

Step 3: Count the number of patients

Total number of patients (n) = 5

Step 4: Calculate the average (mean)

Applying the formula:

$$\text{Average Age } (\bar{x}) = \Sigma x / n = 229 / 5 = 45.8 \text{ years}$$

Result: The average age of the patients in this dataset is 45.8 years. This means that if I were to distribute the ages evenly among all five patients, each would be 45.8 years old.

Calculation 2: Average Heart Rate

Step 1: Identify all heart rate values from the dataset

From the five patient records, I extract the heart rate values (measured in beats per minute):

Patient 1001: 72 bpm Patient 1002: 78 bpm Patient 1003: 88 bpm Patient 1004: 76 bpm Patient 1005: 70 bpm

Step 2: Add all heart rate values together

Sum of heart rates (Σx) = 72 + 78 + 88 + 76 + 70

= 150 + 88 + 76 + 70 (adding first two numbers) = 238 + 76 + 70 (adding next number) = 314 + 70
70 (adding next number) = 384 bpm (final sum)

Step 3: Count the number of patients

Total number of patients (n) = 5

Step 4: Calculate the average (mean)

Applying the formula:

$$\text{Average Heart Rate } (\bar{x}) = \Sigma x / n = 384 / 5 = 76.8 \text{ bpm}$$

Result: The average heart rate of the patients in this dataset is 76.8 beats per minute. This represents a typical resting heart rate for this group of patients during their clinic visits.

Summary of Results

The statistical analysis of the five-patient healthcare dataset yielded the following results:

- 1. **Average Age:** 45.8 years
- 2. **Average Heart Rate:** 76.8 beats per minute

These results are summarized in the table below:

Variable	Sum of Values	Average (Mean)
Age (years)	229	45.8
Heart Rate (bpm)	384	76.8

Interpretation

The calculated averages provide valuable insights into the patient population:

Age Distribution: The average age of 45.8 years indicates a middle-aged patient population. The individual ages range from 28 to 62 years, showing a fairly diverse age distribution with a span of 34

years between the youngest and oldest patients. Understanding age distribution is critical in healthcare analytics as it helps identify demographic trends and potential age-related health concerns (Rothman, Greenland, & Lash, 2008).

Heart Rate Analysis: The average heart rate of 76.8 bpm falls within the normal resting heart rate range for adults (60-100 bpm) as defined by the American Heart Association (2023). The individual heart rates ranged from 70 to 88 bpm, with the highest reading coming from the patient seeking urgent care for flu symptoms, which is expected as illness can elevate heart rate. This variation demonstrates the importance of considering clinical context when interpreting vital signs data.

Conclusion

This analysis demonstrates the fundamental process of calculating descriptive statistics from healthcare data. By computing the average age (45.8 years) and average heart rate (76.8 bpm), I gain a basic understanding of the central tendency of these important numerical variables in the patient population. These calculations form the foundation for more advanced statistical analyses that could be performed on larger healthcare datasets to identify trends, patterns, and potential areas of clinical concern (Sullivan, 2012).

The step-by-step methodology presented here, identifying values, summing them, counting observations, and dividing to find the mean, is a universally applicable approach for calculating averages across any numerical dataset in healthcare or other domains. As healthcare increasingly relies on data-driven decision-making, proficiency in basic statistical analysis becomes essential for healthcare professionals (Kaplan & Haenlein, 2019).

References

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